

Ph.D. Qualifying Exam 2022: Statistical Inference

July 20, 2022
10 AM to 2 PM

Rules: This is a four hour exam consisting of two problems, both of which should be answered. You may use ONE page (front and back, 8.5 by 11 inch paper) HANDWRITTEN “information sheet” for EACH COURSE (Stat 593 and Stat 594) on which this exam is based. No other references are allowed. No use of cell phones, calculators, computers, or other electronic devices is permitted for any reason during the time while the exam is being administered, including any time during which you leave the exam room for any reason. **Show your work and justify your answers clearly. If you use known results (e.g., results stated/proven in class), please state them clearly.** Please put your name or initials on every page. Finally, write your answers clearly and legibly on blank papers, preferably using pens, leaving an impression deep enough for the papers to be scanned. At the end of the exam, separate out the bunches of pages for problems 1 and 2, and hand them over to the proctor. Good luck!

Problem 1

PART - I

Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be an iid sample from a bivariate normal $N(0, 0, \sigma_1^2, \sigma_2^2, \rho)$ distribution, i.e., the marginal means are zero, the marginal variances are $\text{var}(X_j) = \sigma_1^2$ and $\text{var}(Y_j) = \sigma_2^2$, and the correlation coefficient is $\rho = \text{cor}(X_j, Y_j)$ for $j = 1, \dots, n$. Let $\boldsymbol{\theta} = (\sigma_1^2, \sigma_2^2, \rho)$ denote the parameter vector of interest.

Thus the joint PDF of each (X_j, Y_j) is

$$\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left(\frac{X_j^2}{\sigma_1^2} + \frac{Y_j^2}{\sigma_2^2} - \frac{2\rho X_j Y_j}{\sigma_1\sigma_2}\right)\right)$$

(1) Find a complete sufficient statistic for $\boldsymbol{\theta}$.

(2) Show that the MLE of $\boldsymbol{\theta}$ is

$$\hat{\boldsymbol{\theta}} = (\hat{\sigma}_1^2, \hat{\sigma}_2^2, \hat{\rho}) = \left(\frac{1}{n} \sum_{j=1}^n X_j^2, \frac{1}{n} \sum_{j=1}^n Y_j^2, \frac{\sum_{j=1}^n X_j Y_j}{\sqrt{\sum_{j=1}^n X_j^2 \sum_{j=1}^n Y_j^2}}\right),$$

(3) Show that the statistics $\hat{\sigma}_1^2$, $\hat{\sigma}_2^2$, and $\hat{\rho}$ in (2) are mutually independent if $\rho = 0$.

(4) Show that the Likelihood Ratio Test for $H_0 : \rho = 0$ vs. $H_1 : \rho \neq 0$ amounts to saying “reject H_0 when $\hat{\rho}^2$ is large”.

PART II

For the remaining part of this problem, assume $\sigma_1^2 = \sigma_2^2 = 1$, that is, the marginal distributions of X and Y are both standard normal. Only the correlation coefficient ρ is the unknown parameter.

(5) Find the score function $S(\rho, X, Y)$ for a single pair of observations (X, Y) .

- (6) Find the expected Fisher information $I_1(\rho)$ for a single pair of observations (X, Y) .
- (7) Show that the maximum likelihood estimator (MLE) of ρ can be obtained as a solution of a cubic equation $h(\hat{\rho}_{\text{MLE}}) = 0$.
- (8) Write down the Newton-Raphson iteration for solving the ML equation.
- (9) Write down the Fisher scoring iteration for solving the ML equation.
- (10) Obtain the asymptotic distribution of the MLE and use it to obtain a large-sample confidence interval for ρ with confidence coefficient $1 - \alpha$ (you may use a plug-in approach).
- (11) Now consider the estimator:

$$\hat{\rho}_2 = \frac{1}{n} \sum_{i=1}^n X_i Y_i,$$

obtained by substituting ones for the sample variances $\sum_{j=1}^n X_j^2/n$ and $\sum_{j=1}^n Y_j^2/n$ in the expression for the sample correlation coefficient $\hat{\rho}$ in (2). Does this estimator have any desirable property that $\hat{\rho}_{\text{MLE}}$ does not? Derive the asymptotic distribution of this estimator and state whether for large samples you would prefer this estimator over the estimator $\hat{\rho}_{\text{MLE}}$ in (7). Explain your answer. (Recall that for $X \sim N(0, 1)$, $E(X^4) = 3$.)

Problem 2

Suppose that the training data, (x_i, y_i) , $1 \leq i \leq n$, are independent and identically distributed observations as (x, y) , satisfying a linear regression model

$$y = x^\top \beta^* + \varepsilon,$$

where x is a $p \times 1$ covariate vector, y is a response variable, β^* is a $p \times 1$ vector of unknown parameters, and ε is a noise variable, independent of x . In addition, suppose that the following conditions hold:

(A1) ε is sub-Gaussian with parameter σ : $P(|\varepsilon| \geq t) \leq 2e^{-t^2/(2\sigma^2)}$ for any $t > 0$.

(A2) $|x| \leq B_0$ almost surely for a constant $B_0 > 0$.

Let $\hat{\beta}$ be a Lasso estimator of β^* :

$$\hat{\beta} = \operatorname{argmin}_{\beta \in \mathbb{R}^p} \{\ell_n(\beta) + \lambda \|\beta\|_1\}, \quad (1)$$

where $\ell_n(\beta) = \frac{1}{2n} \sum_{i=1}^n (y_i - x_i^\top \beta)^2$, $\|\beta\|_1$ is the L_1 norm of β , and $\lambda > 0$ is a (pre-specified) tuning parameter. Let (x_0, y_0) be another observation, independent and identically distributed as the training data. The questions are designed to study the out-of-sample prediction error:

$$E_{\text{out}}\{(y_0 - x_0^\top \hat{\beta})^2\}, \quad (2)$$

where $E_{\text{out}}(\cdot)$ is taken with respect to (x_0, y_0) , *conditionally on the training data* $\{(x_i, y_i) : i = 1, \dots, n\}$. Hence (2) should be considered a function of the training data. For $\delta > 0$, denote $\lambda_0 = \sqrt{\frac{\log(p/\delta)}{n}}$, with dependency on δ suppressed in the notation.

For each question (i)-(v), you may use the results from the preceding questions, whether or not you have answered those questions.

(i) Show that

$$E_{\text{out}}\{(y_0 - x_0^\top \hat{\beta})^2\} = E_{\text{out}}\{(y_0 - x_0^\top \beta^*)^2\} + E_{\text{out}}\{(x_0^\top (\hat{\beta} - \beta^*))^2\}.$$

(ii) Denote $\Sigma = E(xx^T)$ and $\tilde{\Sigma} = n^{-1} \sum_{i=1}^n x_i x_i^T$. For any sufficiently small $\delta > 0$, show that with probability $1 - k_1 \delta$ for some constant $k_1 > 0$, we have

$$\|\tilde{\Sigma} - \Sigma\|_{\max} \leq B_1 \lambda_0, \quad (3)$$

where $B_1 > 0$ is a constant, depending on B_0 and $\|A\|_{\max} = \max_{1 \leq j, k \leq p} |A_{jk}|$ for any matrix A with elements A_{jk} , $1 \leq j, k \leq p$. Denote as Ω_1 the event that (3) holds.

(iii) Suppose that the following conditions hold:

(A3) $b^T \Sigma b \geq \nu_0^{*2} (\sum_{j=1}^p b_j^2)$ for any $b = (b_1, \dots, b_p)^T \in \mathbb{R}^p$, where $\nu_0^* > 0$ is a constant.

(A4) $|S_{\beta^*}| \lambda_0 \leq \eta_0$ for a constant $\eta_0 > 0$, where S_{β^*} denotes the set of nonzero elements of β^* and $|S_{\beta^*}|$ denotes the size of S_{β^*} .

Show that in the event Ω_1 ,

$$\nu_0^2 \left(\sum_{j \in S_{\beta^*}} |b_j| \right)^2 \leq |S_{\beta^*}| (b^T \tilde{\Sigma} b), \quad (4)$$

for any $b = (b_1, \dots, b_p)^T \in \mathbb{R}^p$ satisfying $\sum_{j \notin S_{\beta^*}} |b_j| \leq \xi_0 \sum_{j \in S_{\beta^*}} |b_j|$, where $\nu_0 > 0$ and $\xi_0 > 1$ are constants, depending on ν_0^* and η_0 . Note that (4) involves $\tilde{\Sigma}$, not Σ .

(iv) Assume that conditions (A1)–(A4) hold. Formulate an event Ω_2 such that $P(\Omega_2) \geq 1 - k_2 \delta$ for some constant $k_2 > 0$, and in the event $\Omega_1 \cap \Omega_2$, the following result holds for some suitable choice of λ in (1):

$$\frac{1}{n} \sum_{i=1}^n (x_i^T (\hat{\beta} - \beta^*))^2 \leq M_0 |S_{\beta^*}| \lambda_0^2, \quad (5)$$

$$\|\hat{\beta} - \beta^*\|_1 \leq M_0 |S_{\beta^*}| \lambda_0, \quad (6)$$

where $M_0 > 0$ is some constant (independent of n, p, δ). Please carefully prove $P(\Omega_2) \geq 1 - k_2 \delta$ for your choice of Ω_2 , but do not need to prove (5)–(6).

(v) For the choice of λ in Question (iv), show that in the event $\Omega_1 \cap \Omega_2$,

$$E_{\text{out}} \{(y_0 - x_0^T \hat{\beta})^2\} \leq E_{\text{out}} \{(y_0 - x_0^T \beta^*)^2\} + M_1 |S_{\beta^*}| \lambda_0^2,$$

where $M_1 > 0$ is some constant (independent of n, p, δ).